http://goo.gl/O359s
Intel's OASIS technology creates an interactive surface anywhere. Seen here is a representation of a kitchen table

G-speak
John Underkoffler demos G-speak, the Minority Report UI. Go to **http://goo.gl/svTT**

base which has myriad 40 x 40 pixel digital models of the hand clad in the glove in a range of positions. Since lycra is a material that can be easily stretched, depending on the user's hand size, system needs calibration. The user has to place an 8.5-inch by 11-inch piece of paper on a flat surface in front of the webcam and press his or her hand against it. Within three seconds of doing that the system is calibrated.

5 KINECT HACKS YOU SHOULD CHECK OUT

http://goo.gl/lgczl
Play Mario using Kinect
http://goo.gl/HIzAm
Controlling Lights via kinect
http://goo.gl/3Nvub
Using Kinect sensor in 3D enviornment recreation
http://goo.gl/wygIQ
Hollow Man revisited
http://goo.gl/7zGcc
Interactive puppy prototype with Kinect

The immediate application of this would be in gaming systems. But in a video demonstration Wang has shown the use of this system in 3D modeling. It can also be used by mechanical engineering students to study cross sections of machines by analyzing individual parts. Wang has further created a patchwork shirt which can detect the upper body movement with the help of two cameras.

Gestures on Mobile phones beyond Touch

The iPhone has revolutionized the way we interact with our phones now, ever since Apple introduced the multi-touch gesture on a touchscreen phone. We have seen lots of improvements in that department, with a lot of companies coming out with more touchscreen phones now, than ever before. Apart from voice commands, ever thought of how navigation on your cellphone would be like without touching it in any manner?

A team at Dartmouth college has developed a technology that will not require you to use your hands to operate your phones basic functionalities.

The EyePhone, as the name implies, uses eye tracking and eye actions such as winking to register inputs. This can work with any smartphone having a front facing camera. The system tracks the user's eye movement around the phone's display using the phone's front facing camera and detects the eye-blink to activate the target application. The algorithm developed by the team comprises four stages like eye detection phase, an open eye template creation, an eye-tracking phase and a blink detection phase. This system can be used in applications like car driver safety, whereby the eye phone can be used to detect drivers drowsiness and distraction while driving and also for hands free navigation for quick tasks like accepting or rejecting calls, going to the music menu or messaging menu, etc.

An Israeli company eyeSight specializing in touch-free interfaces for portable devices has a solution for operating your smartphone without touching it, by using hand gestures. The said phone needs a front facing camera to identify the gestures. The solution – comprising of advanced image processing algorithms – tracks the gestures and converts them to input commands which are then used to control the phone and its applications. A simple wave of a hand over the phone can silence a ringing call or scroll through photographs or skip tracks in your media player. Now this may seem a round about way of doing things, but if we take this technology beyond cellphones it frees us from using input devices. This technology can be used in areas like personal navigation devices in your cars or operating your TV without a remote control or controlling your laptops volume controls without having to reach the mouse or keyboard. You could simply make a natural dismissive gesture to change the channel if an annoying commercial break comes up or if you are on some TV program that really bores you to death.

This is just a slice of the prospects gesture computing has to offer. In the near future we are going to see a lot of gesture computing based devices around us. Even chip manufacturing giant, Intel has recently announced a new research laboratory called Interaction and Experience Research (IXR) which has set its mission as creating the human input/output interface by the year 2020.

Gesture based computing will not completely replace the input devices like mice, keyboards or touchscreens but will definitely ease up the way in which we interact with machines. Apart from hand gestures, brain controlled devices will become quite prominent in the near future. And even though the current projects make use of one gestural interface at a time, we forsee a future where integration of various types of gestural interfaces like hand gestures, eye tracking will be the norm. This will just bring us more closer to our machines or better still, make machines more human, whichever way you look at it.

GESTURE CUBE

Imagine arranging five iPad's in the form of a cube (four sides and one top) and each of the iPad capable of communicating with the adjoining one, with inputs taken from your hand gestures. This is the concept that Germany based company Ident Technology AG is working on. It is called Gesture Cube and it has five screen arranged on the sides and top of a cube. Each screen can interact with its adjoining screen and the user can use hand gestures to activate operations on the screens. This concept was showcased at the Mobile World Congress in Barcelona this year. The screens can show you weather updates, prompt you when a friend sends you a message, can play music and video and recognise devices like cellphones and wirelessly transfer data to and from it.